EXPERIMENT 2:

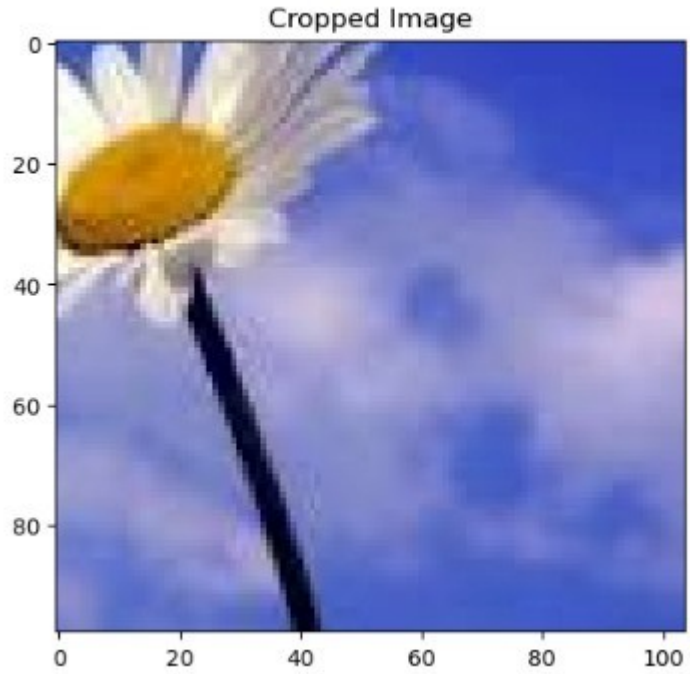
```
import numpy as np import matplotlib.pyplot as
plt img=cv2.imread("Downloads/download.jpg")
img_rgb=cv2.cvtColor(img,cv2.COLOR_BGR2
RGB) plt.imshow(img_rgb) plt.title("Original
Image") plt.axis("off")
```

```
(np.float64(-0.5), np.float64(203.5), np.float64(147.5), np.float64(-0.5))
```

Original Image

```
cropped=img_rgb[50:300 , 100:400]
#cropped_rgb=cv2.cvtColor(cropped,cv2.COLOR_BGR2
RGB) plt.imshow(cropped) plt.title("Cropped Image")
#plt.axis("off")
```

```
Text(0.5, 1.0, 'Cropped Image')
```



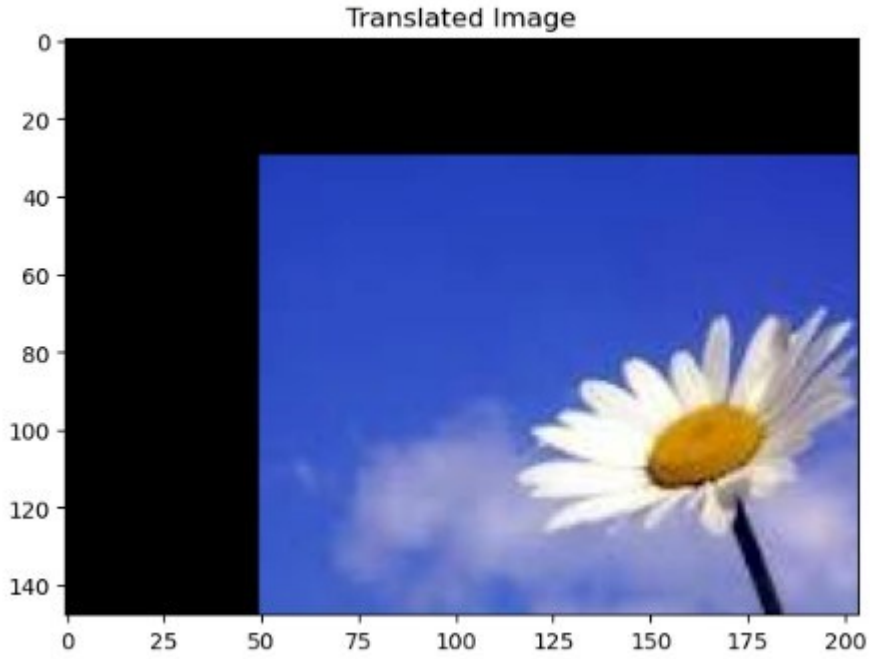

```
resized=cv2.resize(img_rgb,(300,
300)) plt.imshow(resized)
plt.title("Resized Image")
plt.axis("off")
```

```
(np.float64(-0.5), np.float64(299.5), np.float64(299.5), np.float64(-0.5))
```

```
rows,cols=img.shape[:2]
M=np.float32([[1,0,50],[0,1,30]])
translated=cv2.warpAffine(img,M,(cols,rows))
translated_rgb=cv2.cvtColor(translated,cv2.COLO
R_BGR2RGB) plt.imshow(translated_rgb)
plt.title("Translated Image")
```

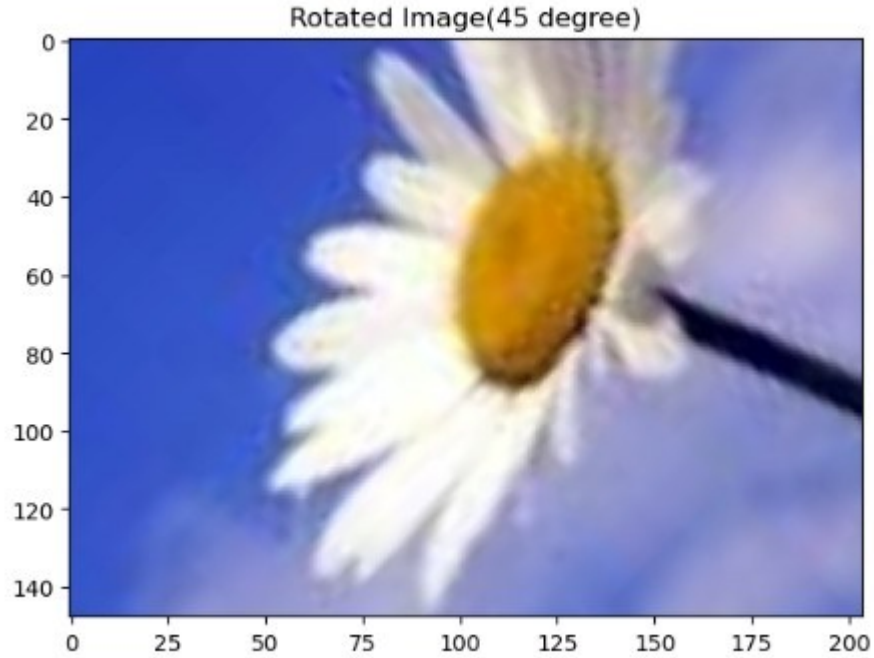
#plt.axis("off")

```
Text(0.5, 1.0, 'Translated Image')
```



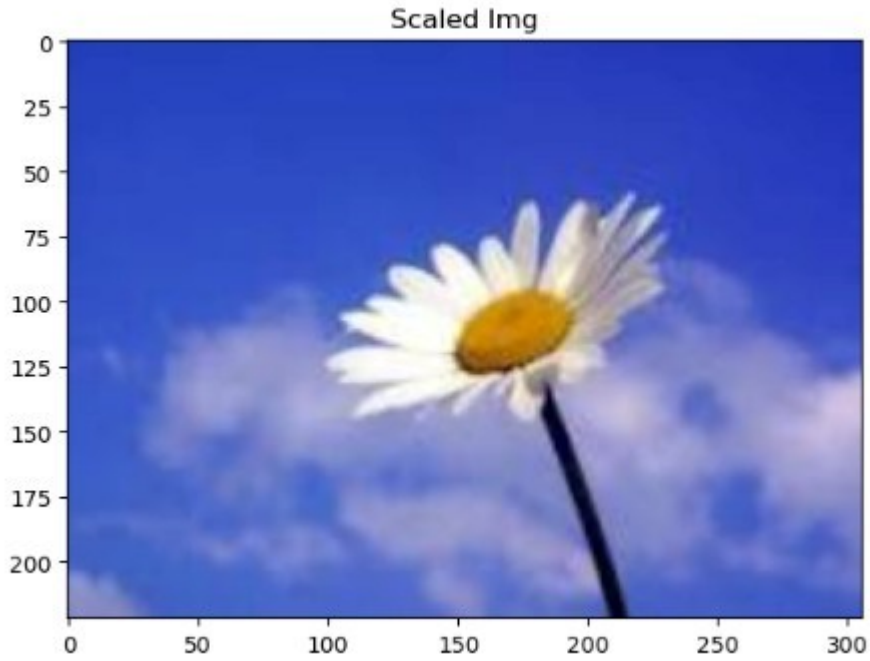

center=(cols//2,rows//2) #floor division
M=cv2.getRotationMatrix2D(center,45,2.0)
rotated=cv2.warpAffine(img,M,(cols,rows))
rotated_rgb=cv2.cvtColor(rotated,cv2.COLOR_BGR2RGB)
plt.imshow(rotated_rgb) plt.title("Rotated Image(45 degree)")

```
Text(0.5, 1.0, 'Rotated Image(45 degree)')
```



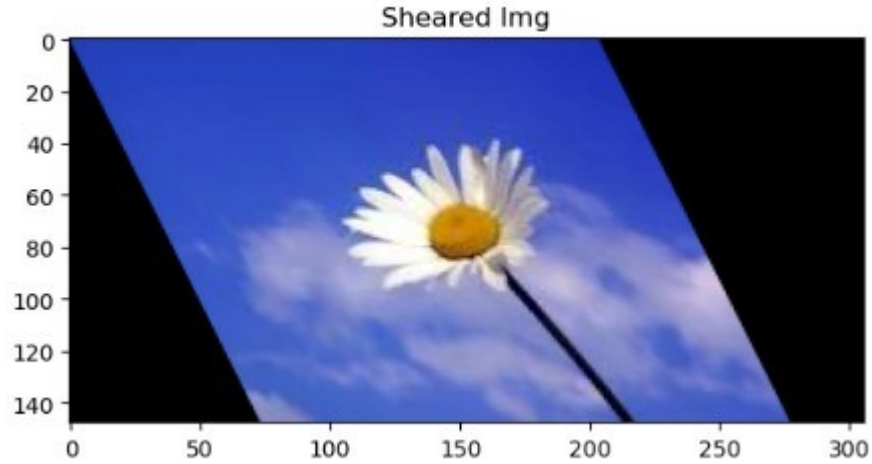

scaled = cv2.resize(img,None,fx=1.5,fy=1.5)
scaled_rgb=cv2.cvtColor(scaled,cv2.COLOR_BGR2
RGB) plt.imshow(scaled_rgb) plt.title("Scaled Img")

```
Text(0.5, 1.0, 'Scaled Img')
```



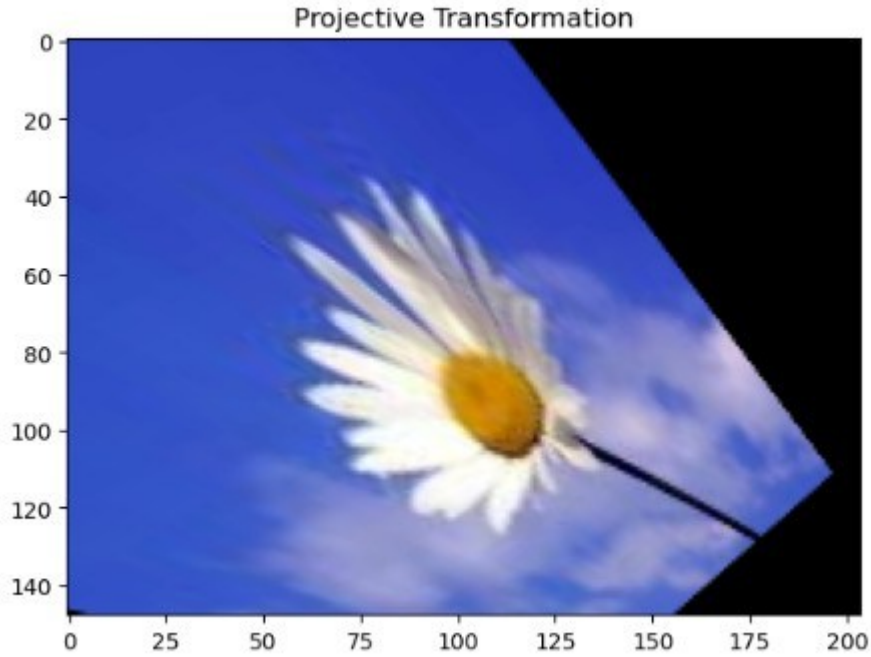

```
rows,cols=img.shape[:2]
M=np.float32([[1,0.5,0],[0,1,0]])
sheared=cv2.warpAffine(img,M,(int(cols*1.5),rows
))
sheared_rgb=cv2.cvtColor(sheared,cv2.COLOR_
BGR2RGB) plt.imshow(sheared_rgb)
plt.title("Sheared Img")
```

```
Text(0.5, 1.0, 'Sheared Img')
```




```
flipped=cv2.flip(img,1)
flipped_rgb=cv2.cvtColor(flipped,cv2.COLOR_BGR2
RGB) plt.imshow(flipped_rgb) plt.title("Reflected
Img(horizontal flip)")
```

```
Text(0.5, 1.0, 'Reflected Img(horizontal flip)')
```

```
rows,cols=img.shape[:2]
pts1=np.float32([[50,50],[cols-50,50],[50,rows-
50],[cols-50,rows-50]])
pts2=np.float32([[10,100],[cols-100,50],[100,rows-
10],[cols-50,rows-50]])
M=cv2.getPerspectiveTransform(pts1,pts2)
perspective=cv2.warpPerspective(img,M,(cols,row
s))
perspective_rgb=cv2.cvtColor(perspective,cv2.CO
LOR_BGR2RGB) plt.imshow(perspective_rgb)
plt.title("Projective Transformation")
```

```
Text(0.5, 1.0, 'Projective Transformation')
```